

ece4-exp

EC-Earth4 Experiment Configuration Made Simple

From hours of YAML editing to 30 seconds

The Problem

Creating an EC-Earth4 experiment config is painful:

-  Edit 257-line YAML file manually
-  Calculate node layouts and processor distribution
-  Match components, grids, platform paths
-  Easy to make mistakes (typos, incompatible combinations)
-  Repeat for every experiment

Result: 2-4 hours for experienced users, longer for newcomers

The Solution

One command generates production-ready configs:

```
ece4-exp generate gcm-sr 10 a001
```

That's it. You get:

-  Correct component configuration (OIFS + NEMO + XIOS + OASIS)
-  Calculated node layout (10 nodes × 112 cores = 1120 procs)
-  Base config fetched from upstream EC-Earth4 repo
-  Validated and ready to run

Time saved: ~3.5 hours per experiment

Pre-Built Recipes

Ready-to-use experiment templates:

Recipe	Description	Nodes*	Components
gcm-sr	Coupled atmosphere-ocean	10	OIFS + NEMO + XIOS + OASIS
omip-sr	Ocean-only with ERA5 forcing	2	NEMO + XIOS
amip-sr	Atmosphere-only (prescribed SST)	8	OIFS + XIOS + AMIPFR
ccycle-sr	Carbon cycle (with LPJG)	10+	OIFS + NEMO + LPJG + XIOS

*MareNostrum5 (112 cores/node). All at standard resolution: TL255L91 + eORCA1L75.

All recipes: Tested, validated, production-ready

How It Works

YAML overlay system - configurations merged in order:

```
1. Base config      (fetched from EC-Earth4 git repo, pinned version)
   ↓
2. Platform         (node layout, SLURM settings for your HPC)
   ↓
3. Recipe           (experiment type: coupled, ocean-only, ...)
   ↓
4. Your defaults    (~/.config/ece4-exp/defaults.yml)
   ↓
5. CLI flags        (per-run overrides)
   ↓
Generated config (a001_experiment.yml, ~229 lines)
```

You only write what's **different from the defaults**.

The tool handles the rest: node math, platform paths, component config.

Real Examples

Coupled atmosphere-ocean (10 nodes):

```
ece4-exp generate gcm-sr 10 a001  
# → OIFS + NEMO + XIOS + OASIS + RNFM  
# → TL255L91 + eORCA1L75
```

Ocean-only (2 nodes):

```
ece4-exp generate omip-sr 2 o001  
# → NEMO + XIOS (no atmosphere)  
# → Forced by ERA5
```

Different science, different resources - **same simple command.**

Setup: Once and Forever

One-time installation (2 minutes):

```
pip install ece4-exp
```

Configure your defaults (1 minute):

```
ece4-exp setup  
# Edit ~/.config/ece4-exp/defaults.yml
```

Set your platform, account **once**.

Then just:

```
ece4-exp generate RECIPE NODES EXPID
```

Override when needed: `--platform ecmwf-hpc2020`

Advanced Features

Create custom recipes:

```
# Generate and tweak
ece4-exp generate gcm-sr 10 test
vim test_experiment.yml

# Save as new recipe
ece4-exp save --expid test --recipe gcm-sr

# Reuse it
ece4-exp generate my-recipe 10 a001
```

Scripting support:

```
# Quiet mode (no colors, for scripts)
ece4-exp generate gcm-sr 10 a001 --quiet
```

Platform Support

Currently supported:

-  BSC MareNostrum 5
-  ECMWF HPC2020

Adding a new platform:

1. Create `~/.config/ece4-exp/platforms/<name>.yaml`
2. Set `ppn` (cores per node) and node layouts per experiment type
3. Done!

Example (key fields):

```
ppn: 128                # Cores per node
slurm-wrapper-taskset:
  CPLD-SR:              # Coupled GCM layout
  slurm:
```

Get Started

The full workflow:

```
pip install ece4-exp
ece4-exp setup                # configure once
ece4-exp generate gcm-sr 10 a001 # generate config
ece4-exp deploy a001          # push to HPC
```

Then on the HPC:

```
cd $scratch/ecearth4/scripts/runtime
se user.yml platform.yml a001_experiment.yml scriptlib/main.yml
```

Documentation:

- `ece4-exp --help` — Command reference
- <https://ece4-exp.readthedocs.io>